

Constantinos Xenophontos

Curriculum Vitae



Postdoctoral researcher, IRCCS Humanitas Research Hospital, Milan
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Employment

Since 2023 **Postdoctoral researcher**, IRCCS Humanitas Research Hospital, Milan, Italy
PI: Dr. Gabriele Micali, Microbial Ecology Lab

Education

2016 – 2020 **PhD researcher**, Friedrich Schiller University Jena & German Centre for Integrative Biodiversity Research (iDiv) Halle-Jena-Leipzig, Germany
Dissertation: *Testing ecological theories with microbial communities*.
Supervisors: Prof. Dr. Kirsten Küsel, Prof. Dr. W. Stanley Harpole.

2014 – 2015 **MRes Ecology, Evolution & Conservation (Merit)**, Imperial College London, UK
Thesis 1: *Variation in behaviour and resistance to fungal endoparasites between closely related clones of the anciently asexual rotifer Adineta vaga*.
Thesis 2: *Effects and interactions on phenotypic trajectories and metabolic activity of community bacteria*.

2010 – 2014 **BSc (Hons) Biology (2:1)**, University of Hull, UK
Team research project in ecology, DNA barcoding and bioinformatics: *Is soil mesofauna abundance and diversity related to farming methods?*

Publications

Xenophontos, C., Harpole, W. S., Küsel, K., & Clark, A. T. (2022). Cheating promotes coexistence in a two-species one-substrate culture model. *Frontiers in Ecology and Evolution*, 9, 786006. <https://doi.org/10.3389/fevo.2021.786006>

Xenophontos, C., Taubert, M., Harpole, W. S., & Küsel, K. (2021). Phylogenetic and metabolic diversity have contrasting effects on the ecological functioning of bacterial communities. *FEMS Microbiology Ecology*, 97(3), fiab017. <https://doi.org/10.1093/femsec/fiab017>

The 2021 *FEMS Microbiology Ecology* article was selected as an Editor's Choice.

Conference presentations

2026 May **Microfluidic Horizons 2026**, oral presentation
Stronger at the seams: antibiotic stress enhances amino acid leakage and cross-feeding.
Xenophontos C., Vaidyanathan A., Micali G.

- 2025 Dec **BES Annual Meeting 2025**, poster presentation
Stronger at the seams: antibiotic stress enhances amino acid leakage and cross-feeding. Xenophontos C., Vaidyanathan A., Micali G.
- 2025 Jul **FEMS Micro 2025**, oral and poster presentations
Stronger at the seams: antibiotic stress enhances amino acid leakage and cross-feeding. Xenophontos C., Vaidyanathan A., Micali G.
- 2020 Dec **ISME Virtual Microbial Ecology Summit**, poster presentation
Coexisting with cheaters: microbial cheating as a snowdrift game model. Xenophontos C., Harpole W.S., Küsel K., Clark A.T. doi:10.13140/RG.2.2.36728.26886
- 2019 Dec **BES Annual Meeting 2019**, oral presentation
Phylogenetic and functional dissimilarity have contrasting effects on the ecological functioning of synthetic bacterial communities. Xenophontos C., Taubert M., Harpole W.S., Küsel K. doi:10.13140/RG.2.2.16123.67361
- 2019 Sep **SAME16**, poster presentation
Species phylogenetic and functional diversity contribute to ecological functioning of groundwater bacterial communities. Xenophontos C., Taubert M., Harpole W.S., Küsel K.
- 2019 Aug **5th iDiv Conference**, poster presentation
Species phylogenetic and functional diversity contribute to ecological functioning of groundwater bacterial communities. Xenophontos C., Taubert M., Harpole W.S., Küsel K.
- 2018 Dec **4th iDiv Conference**, oral presentation
Species phylogenetic and functional diversity contribute to ecological functioning of groundwater bacterial communities. Xenophontos C., Taubert M., Harpole W.S., Küsel K.
- 2018 Aug **ISME17**, poster presentation
Species phylogenetic and functional diversity contribute to ecological functioning of groundwater bacterial communities. Xenophontos C., Taubert M., Harpole W.S., Küsel K.
- 2017 Sep **3rd iDiv Conference**, poster presentation
Shifts in ecological functioning of groundwater bacteria driven by species interactions. Xenophontos C., Geesink P., Kumar S., Wegner C.-E., Harpole W.S., Küsel K. doi:10.13140/RG.2.2.24728.49929

Teaching and supervision

- Since 2023 Co-supervision of PhD and MSc students with Dr. Micali (lab work and data analysis), IRCCS Humanitas Research Hospital
- 2019 May Soil DNA extraction and PCR practical course, Friedrich Schiller University Jena

Academic service

- 2026 **Microfluidic Horizons 2026**, organising committee
Social media and communications; session moderator

Outreach

2023	<i>Expo per lo Sport</i> , public event, Humanitas booth, Milan DNA-extraction instructor for children
2022	<i>Holobiontinnen</i> , art performance exhibit, Leipzig (LOFFT, Ballhaus Ost) Consulting microbiologist
2019	<i>Long Night of Sciences</i> , public event, Jena Instructor, microbiology art for children

Grants, awards and studentships

2023	EMBO Postdoctoral Fellowship, shortlisted and interviewed; not funded
2023	HFSP Long-Term Fellowship, applied; not funded
2019	SAME16 conference travel grant
2016	yDiv Doctoral Programme studentship, German Centre for Integrative Biodiversity Research (iDiv) Halle-Jena-Leipzig

Societies

- International Society for Microbial Ecology (ISME), member
- Royal Society of Biology (RSB), member (MRSB)
- British Ecological Society (BES), member
- Federation of European Microbiological Societies (FEMS), affiliate member
- Society for Open, Reliable, and Transparent Ecology and Evolutionary biology (SORTEE), member

Skills

- **Experimental:** bacterial growth assays in 96-well plate readers, microfluidic timelapse microscopy, continuous-culture bioreactors, synthetic bacterial communities, DNA extraction and PCR
- **Computational:** R (tidyverse, Quarto), Python (marimo, Polars), Julia; growth-curve and dose-response analysis; image segmentation and cell tracking (Omnipose); Git and GitHub
- **Hardware and fabrication:** open-source bioreactor firmware (ChiBio); 3D printing and design (PrusaSlicer, BambuStudio, Fusion 360, Meshmixer)